

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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Originating Company	Epic Atlantic Limited
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Specification for Field Gauges

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Revision History

Rev No.	Date	Description of Revision
R01	18.07.22	Issued for Review
R02	07.10.22	Re-issued for review after incorporating TCE comments
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

PG	Pressure Gauge
TG	Temperature Gauge
LG	Level Gauge
RO	Restriction Orifice Plate
RTD	Resistance Temperature Detector

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development of a plant complex comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

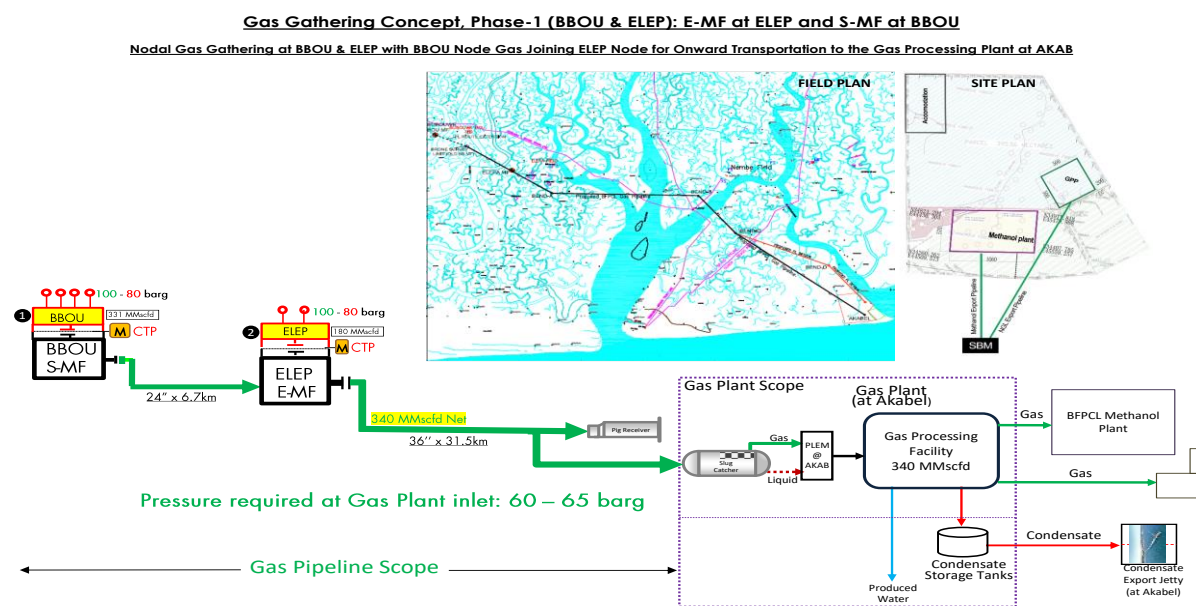


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document defines the basic requirements for the field gauges like Pressure Gauge, Temperature Gauge, RTD and Orifice Plate. The FEED shall fully comply with all specified relevant contractual requirements. This document defines the basic requirements for the engineering and design of the instrumentation for BBOU, ELEM and AKAB pipeline. The purpose of this specification is to ensure a standard approach is maintained to the engineering and design of instrumentation and equipment and that it complies with all relevant national and international standards.

2. Regulations, Codes and Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. General

- 3.1. Field instrumentation shall have a minimum ingress protection of IP65 in accordance with IEC 60529. Ingress protection for Level Gauge Indicator shall be IP-67.

3.2. Materials of Construction

Materials of construction for instrument equipment shall be consistent with corrosion resistance and the pressure and temperature conditions of the process fluids involved. Generally, the parts in contact with the process fluid shall be of a material equal to or better than that used for the related vessel or line. Where instrument design prohibits the use of these materials, or where a suitable material cannot be reasonably procured, the instruments shall be isolated from the process by means of a chemical seal which shall be identified on P&ID.

Wetted parts including sensing elements and also movement parts shall be of AISI 316 stainless steel. The material employed in special services shall be selected according to application from SS316L, Titanium, Monel, Hastelloy and etc. The materials of wetted part in sour services shall generally be selected and treated in accordance with the recommendation of NACE MR 0103

The following materials shall not be used in contact with the product:

Aluminium for wetted parts, Zinc, Asbestos, Brass, Bronze, Copper. Heavy metals including Mercury, Antimony and Cadmium. Plastics containing free Phenol, formaldehyde or plasticizers.

3.3. Pressure & temperature rating

Pressure and temperature rating of instruments and components shall conform to the design rating of the process system to which they are connected. Instruments which can be exposed to vacuum shall have under-range protection to full vacuum.

4. Pressure Gauge

4.1. Pressure gauges- Basic Requirement

- 4.1.1. Pressure gauge accuracy shall be + 1.0% of full range.
- 4.1.2. Dial size shall be 150 mm and cases shall be stainless steel screw on or bayonet bezel type. Blowout disc protection shall be provided and gauges shall be orientated such that they vent safely. All pressure gauges shall be supplied with Solid Fronts.
- 4.1.3. Gauge windows shall be constructed from safety pattern / toughened glass.
- 4.1.4. Gauges shall be normally Bourdon tube type. Bellows or diaphragm-actuated gauges shall be used for lower pressure ranges. Chemical seals with body and diaphragm in stainless steel shall be used if the medium tends to crystallise, is highly viscous, or where suspended matter exists. The process connection of flanged pressure gauge should be 2" flange.
- 4.1.5. The seal fluid wherever specified shall normally be silicone oil. For higher temperature, vendor shall suggest the seal fluid compatible with process temperature.
- 4.1.6. Bourdon tube material shall be type 316 stainless steel, as a minimum, with stainless steel movement, except where process conditions require more stringent materials. Martensitic stainless steel must be used for gases containing hydrogen.
- 4.1.7. Over-range protection shall be 1.3 times the maximum scale range. Where a gauge is subject a greater pressure, a gauge saver shall be used. All pressure gauges installed on the Pig Launcher and Pig Receiver shall have Gauge Saver.
- 4.1.8. A gauge valve shall be provided for isolation and calibration purposes. A primary isolation valve shall be provided where shown in the P&ID.
- 4.1.9. Gauges shall be connected to the piping by means of Mono-flanges.
- 4.1.10. Gauge connections shall be 1/2" NPTM.

4.2. Design

- 4.2.1. Pressure gauges shall consist of a sensing element integral with a case and indicating mechanism.
- 4.2.2. For Chemical seal type gauge, the diaphragm design shall be welded type. The top chamber shall be exactly contoured to fit diaphragm and thus eliminating distortion of diaphragm. The top and bottom chamber shall be held together by clamping bolts and makes the assembly doubly sealed. The bottom chamber should be easily removable for purpose of cleaning, without refilling or recalibrating the protected instruments. Capillary tubing extension length, when required shall

be as indicated on the data sheet. The capillary material shall be of stainless steel and shall be with SS armouring.

- 4.2.3. A rotary geared movement shall be used for all pressure gauges. Movement must have an extremely low co-efficient of friction, resist abrasions and must be capable of withstanding the most severe conditions of vibration, pulsation and corrosion.
- 4.2.4. The case shall have bayonet bezel or front threaded ring with gasket and shatterproof glass and a rear safety blow out disc.
- 4.2.5. For vacuum measurement stopper shall be provided at the maximum vacuum range on dial.

4.3. Functional Requirements

- 4.3.1. Range shall be as indicated against each tag. Where the exact range is not available, the vendor shall offer his standard range which most closely approximates that specified so that the operating pressure is indicated within 50 to 70% of the scale.
- 4.3.2. All gauges must be capable of withstanding over-pressures equivalent to 130% of maximum scale reading.
- 4.3.3. For gauges operating in full vacuum/subject to design condition of full vacuum an under range protection shall be provided.
- 4.3.4. Absolute pressure element shall be fully compensated for changes in barometric pressure.
- 4.3.5. Range adjustment shall be achieved mechanically and internally with linkages or pointer shift.
- 4.3.6. Zero adjustment shall be achieved mechanically and externally with a pointer shift screw.

4.4. Tag List-Pressure Gauges

Sr. No.	Tag No.	Line Rating	Pressure Range	Instrument Range	Accessories
1	BBOU-PI-1001	900# RF	0-110 Barg	0-160 barg	Solid Front
2	BBOU-PI-1002	900# RF	0-110 Barg	0-160 barg	Solid Front
3	BBOU-PI-1010	900# RF	0-110 Barg	0-160 barg	Solid Front
4	BBOU-PI-1011	900# RF		0- 1 barg	Solid Front, gauge saver
5	BBOU-PI-1012	900# RF	0-110 Barg	0-160 barg	Solid Front
6	ELEP-PI-1001	900# RF	0-110 Barg	0-160 barg	Solid Front
7	ELEP-PI-1002	900# RF	0-110 Barg	0-160 barg	Solid Front
8	ELEP-PI-1010	900# RF	0-110 Barg	0-160 barg	Solid Front
9	ELEP-PI-1011	900# RF	0-110 Barg	0-160 barg	Solid Front
10	ELEP-PI-1012	900# RF	0-110 Barg	0-160 barg	Solid Front
11	ELEP-PI-1013	900# RF		0- 1 barg	Solid Front, gauge saver
12	ELEP-PI-1014	900# RF	0-110 Barg	0-160 barg	Solid Front
13	ELEP-PI-1015	900# RF		0- 1 barg	Solid Front, gauge saver
14	ELEP-PI-1016	900# RF	0-110 Barg	0-160 barg	Solid Front
15	AKAB-PI-1001	900# RF	0-110 Barg	0-160 barg	Solid Front
16	AKAB-PI-1011	900# RF	0-110 Barg	0-160 barg	Solid Front
17	AKAB-PI-1012	900# RF	0-110 Barg	0-160 barg	Solid Front
18	AKAB-PI-1013	900# RF		0- 1 barg	Solid Front, gauge saver

5. Protecting Thermowells for RTD and TG

- 5.1.1. Thermowell shall be machined from bar stock material. Other materials shall be furnished as required in the piping specification. Temperature elements shall be supplied complete with thermowell, extension, head and integrally mounted transmitter as a complete assembly. All thermowells shall be SS316L and meet the Karman Wake Frequency calculation as per PTC 19.3. If it fails to achieve the wake frequency calculation, collar or vortex shedding design shall be used to meet the velocity criteria. Thermowell design and geometry shall meet the requirements of Part II, Section 3.9 of DEP 31.38.01.24-Gen. At all process conditions, the ratio of the Strouhal frequency and the thermowell natural frequency shall be less than 0.8, as stated in ASME PTC 19.3.
- 5.1.2. Thermowells shall be flanged according to equipment and pipe specifications. The flange size shall be 40 mm (1.1/2 inch) with rating and facing according to the line classification.
- 5.1.3. According to API RP 551, the immersion length of thermowells in the pipes (the length from well tip to pipe wall) shall be maximum 5" or 125mm. The immersion length in vessels liquid shall be 400 mm. In vessels containing gas, the immersion length shall be according to the table below:

Vessel diameter (D)	Thermowell immersion length(mm)
D upto 800 mm	200
$800 < D \leq 1200$ mm	300

The insertion length depends on the length of the nozzle or pipe fitting, taking any possible refractory lining into account. If the thermowell design fails to pass the wake frequency with the give gas velocity even after increasing the tip diameter or other mechanical parameters, vortex shedding type helical thermowell shall be used. Velocity collar shall be avoided.

6. Resistance Temperature Detector (RTD)

6.1. Temperature Elements

- 6.1.1. Where the measuring temperature $\leq 400^{\circ}\text{C}$, 3 wire RTD (Platinum, 100 ohm at 0°C) complying with IEC-60751 Class 'A' shall be used. RTD shall be 6 mm OD, AISI 316 stainless steel sheathed. Test voltage shall be 500V rms. The wire diameter shall be minimum 0.90 mm. All the structure of element shall be spring-loaded.
- 6.1.2. Resistance bulbs for temperature measurement shall be mineral insulated of the three-wire type to compensate for lead wire length. Detectors shall be high speed, high sensitivity type and resistance to vibration and mechanical shocks.
- 6.1.3. Resistance bulb length shall be manufacturer's standard to fit the specified thermowell. High temperature limits, calibration curves and temperature ranges for the resistance bulb shall be in accordance with the following criteria.

Code	Type	Calibration Reference Point Ohms	Curve Temperature Range
PT 100	Platinum	100 at 0°C	-200°C to 650°C
Ni	Nickel	100 at 0°C	-150°C to 290°C
Cu	Copper	10 at 25°C	-150°C to 120°C

- 6.1.4. RTD wiring color coding and Calibration curve for resistance temperature detectors shall be as per IEC 60751 (Latest edition).

6.2. RTD Head Assemblies

- 6.2.1. RTD head assemblies shall consist of a case, a screw-on cover with chain and a terminal block.
- 6.2.2. The case shall be of die cast aluminium weatherproof to minimum IP-65. A heat resistant and moisture proof gasket shall be fitted between the case and the cover. The case shall be connected to the thermowell by means of nipple & union nipple assembly. The material of construction for nipple and union shall be Stainless Steel. Duplex RTD shall be used to avoid downtime. Duplex shall be adopted for all interlock and control temperature measurements and only temperature monitoring it can be single RTD. Both elements of duplex RTD will be connected to one Temperature transmitter. For duplex type RTD, two 1/2" NPT (F) threaded cable entries shall be provided, with one entry plugged.
- 6.2.3. The terminal block shall be suitable for three wire elements. The terminal block assembly shall be moulded and made up of high temperature and high dielectric material.
- 6.2.4. Terminal blocks shall be capable of accommodating wires of at least 1.5 mm². The terminal block shall be readily accessible.
- 6.2.5. The cover chain material shall be corrosion resistant preferably stainless steel.
- 6.2.6. The RTD's located in the hazardous area shall be approved for use in specified area by an authorised testing agency such as ATEX, PTB or equivalent. Evidence of such approval shall be affixed to each equipment and a copy of the certification letter shall be provided.

7. Temperature Gauge

7.1. Local Temperature Indicators

- 7.1.1. Local temperature indicators shall be rigid stem bimetallic dial thermometer type supplied complete with protective thermowells with specification mentioned above. The Bi-metallic Temperature Gauges shall be provided with every Angle type head.
- 7.1.2. Dial size shall be 150 mm nominal and instruments shall be heavy duty 'every angle' head type. Temperature scales shall be direct reading and comply with manufacturer's standard ranges. The Case shall be SS304 and Stem shall be SS316.
- 7.1.3. Scales shall be selected to cover the range requirements for both normal and design conditions of plant operation. Mercury filled systems must not be used.
- 7.1.4. The stem material shall be SS-316 and the case material shall be SS-304.
- 7.1.5. The sensing bulb shall be of SS-316.
- 7.1.6. The filled system with capillary, the material of capillary shall be of SS-316 with SS armouring.
- 7.1.7. Range shall be as per mentioned against each tag. All Temperature gauges shall be factory calibrated for the specified range.
- 7.1.8. All Temperature Gauges shall have a provision for external zero adjustment at the back of the case.
- 7.1.9. All Temperature Gauges shall have over-range protection of 130% of range. Accuracy: $\pm 1\%$ of Span Minimum

7.2. Tag Lists- Temperature Gauges

Sr. No.	Tag No.	Line Rating	Pressure Range	Instrument Range	Accessories
1	BBOU-TW/TI-1001	900# RF	0-110 Barg	0-100 degC	Thermowell
2	BBOU-TW/TE-1001	900# RF	0-110 Barg	0=100 degC	Thermowell
3	BBOU-TW/TE-1003	150#RF	0-10 Barg	0=100 degC	Thermowell
4	ELEP-TW/TI-1001	900# RF	0-110 Barg	0-100 degC	Thermowell
5	ELEP-TW/TE-1001	900# RF	0-110 Barg	0=100 degC	Thermowell
6	ELEP-TW/TI-1003	900# RF	0-110 Barg	0-100 degC	Thermowell
7	ELEP-TW/TE-1003	900# RF	0-110 Barg	0=100 degC	Thermowell
8	ELEP-TW/TE-1003	150#RF	0-10 Barg	0=100 degC	Thermowell
9	AKAB-TW/TI-1001	900# RF	0-110 Barg	0-100 degC	Thermowell
10	AKAB-TW/TE-1001	900# RF	0-110 Barg	0=100 degC	Thermowell

8. Restriction Orifice Plate

8.1. Restriction Orifice Plates

- 8.1.1. Restriction Orifice plates shall be used in the flare line/vent line. Thickness of the plate shall be suitable for pressure condition and sonic velocity consideration of the flare line. Thickness of the Restriction orifice plate shall be calculated based on the R W Miller equation
- 8.1.2. Orifice plates shall have serrations.
- 8.1.3. Material of Construction shall be SS316
- 8.1.4. Restriction orifice plate shall be sharp, square edge, concentric. For restriction orifice plate bevel shall not be provided.
- 8.1.5. Restriction orifice plates shall be sizing with maximum differential pressure of line (100 bar) and flare line which in this project is atmospheric
- 8.1.6. A safety margin of 1.2 times on calculated thickness of the RO plate in the flare line shall be taken.
- 8.1.7. Thickness of any orifice plate, as defined in the R.W.Miller's handbook, depends on the following parameters

Line size

Differential Pressure

Yield Stress of Plate.

β Ratio

Thickness calculation in mm = $t_{min} \geq \sqrt{(0.681 - 0.651\beta)DP / Yieldstress}$ X Pipe I.D.

Pipe ID in mm.

DP in Mpa

Yield Stress in Mpa

8.2. Tag List- Restriction Orifice Plates

Sr. No.	Tag No.	Line Rating	Pressure Range
1	BBOU-RO-1001	900# RF	0-110 Barg
2	BBOU-RO-1003	900# RF	0-110 Barg
3	ELEP-RO-1001	900# RF	0-110 Barg
4	ELEP-RO-1003	900# RF	0-110 Barg
5	ELEP-RO-1004	900# RF	0-110 Barg
6	AKAB-RO-1003	900# RF	0-110 Barg

9. Level Instruments

9.1. Magnetic Level Gauges

- 9.1.1. Magnetic Level gauge shall be used for CBD vessel. Magnetic level gauge shall have IP-67 indication unit.
- 9.1.2. The visible portion of gauge glass shall cover the range of any associated level instrument
- 9.1.3. Magnetic level gauges shall be top or side mounted type and shall consist of a float with extended rod and magnet at the end of extension, magnet tube / chamber and float guide pipe. When the float moves up and down with the liquid level inside the vessel, the magnet moves up and down inside the magnet tube/chamber which in turn operates the indication mechanism.
- 9.1.4. The indication shall consist of bi-colour magnetic rollers mounted on outside the magnet chamber. As the float rises or falls with the liquid level each roller rotates 180 deg and so presents a contrasting colour.
- 9.1.5. The accidental rotation of rollers shall be prevented by the construction/design such as edge magnetisation.
- 9.1.6. Float failure warning shall be provided by mounting the bottom rollers of the indication with their colours reversed. In case float reaches that level, a clear readable, sharp indication will be available.
- 9.1.7. Vendor shall provide magnet for initial correction of bi-colour rollers.
- 9.1.8. Vendor to ensure that readability and dead zone errors for magnetic gauges are minimum and the same shall be highlighted in the offer.
- 9.1.9. Floats shall be designed and manufactured for the specific conditions specified on datasheets. It shall be designed to be adequate for hydrostatic test conditions.
- 9.1.10. Floats shall be hermetically sealed, no vented or pressure equalised construction shall be allowed.
- 9.1.11. The float, magnet tube/ chamber, extension rod, process connection flange, guide pipe shall be of SS316 unless otherwise specified.
- 9.1.12. The guide pipe shall be perforated, suitable for float size and shall be through out the length of float travel.

9.2. Tag Lists-Magnetic Level Gauges

Sr. No.	Tag No.	Line Rating	Pressure Range	Instrument Range(Tentative)	Accessories
1	BBOU-LG-1001	150#RF	-	0-2400 mm	Magnetic Level Gauge
2	ELEP-LG-1001	150#RF	-	0-2400 mm	Magnetic Level Gauge